# Gradient methods for parameter optimization

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## 4.1 Linear neuron for regression: analytical solution

### 4.1.1 Data, Model and Objective

Data:

| | |
|---|---|
| observations | $x \in \mathbb{R}$ |
| label | $y_T \in \mathbb{R}$ |
| Dataset | $\left\{ \left(x^{(1)}, y_T^{(1)}\right), \ldots, \left(x^{(\alpha)}, y_T^{(\alpha)}\right), \ldots, \left(x^{(p)}, y_T^{(p)}\right) \right\}$ |

Model:

linear neuron:

$$y(x; \underline{\mathbf{w}}) = w_0 + w_1 x = \underline{\mathbf{w}}^\top \underline{\mathbf{x}} \tag{4.1.1}$$

with $\underline{\mathbf{x}} := (1, x)^\top \in \mathbb{R}^2$ and $\underline{\mathbf{w}} := (w_0, w_1)^\top \in \mathbb{R}^2$

Cost function:

Quadratic error

$$E^T = \frac{1}{p} \sum_{\alpha=1}^{p} \frac{1}{2} \left( y(\underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}) - y_T^{(\alpha)} \right)^2 \tag{4.1.2}$$

$$= \frac{1}{p} \sum_{\alpha=1}^{p} e^{(\alpha)} \tag{4.1.3}$$

The term $\frac{1}{2}$ is merely for convenience when computing the gradient later.

<u>Optimization</u>:

$$E^T \stackrel{!}{=} \min_{\underline{\mathbf{w}}} \tag{4.1.4}$$

$$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} = \underline{\mathbf{0}} \qquad \text{(satisfied by extrema)} \tag{4.1.5}$$

### 4.1.2 Calculating the gradient

$$\frac{\partial E^T}{\partial \underline{\mathbf{w}}} = \frac{1}{p}\sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} \tag{4.1.6}$$

$$\overset{\text{chain rule}}{=} \frac{1}{p}\sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}}{\partial y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \cdot \frac{\partial y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}{\partial \underline{\mathbf{w}}} \tag{4.1.7}$$

$$= \frac{1}{p}\sum_{\alpha=1}^{p} \left(y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}}) - y_T^{(\alpha)}\right) \cdot \frac{\partial}{\partial \underline{\mathbf{w}}} \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} \tag{4.1.8}$$

using $\frac{\partial}{\partial \underline{\mathbf{a}}}\left(\underline{\mathbf{a}}^\top \underline{\mathbf{b}}\right) = \underline{\mathbf{b}} = (b_1, b_2, \ldots, b_N)^\top$

$$= \frac{1}{p}\sum_{\alpha=1}^{p} \left(\underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - y_T^{(\alpha)}\right) \cdot \underline{\mathbf{x}}^{(\alpha)} \tag{4.1.9}$$

$$= \frac{1}{p}\sum_{\alpha=1}^{p} \left(\underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)}\, y_T^{(\alpha)}\right) \tag{4.1.10}$$

$$=: \underline{\mathbf{g}} \tag{4.1.11}$$

### 4.1.3 Finding minima

Constructing the Hessian matrix $\underline{\mathbf{H}}$ to identify minima among the solutions that satisfy Eq.(4.1.5):

$$\underline{\mathbf{H}} \equiv \frac{\partial^2 E^T}{\partial \underline{\mathbf{w}}^2} = \begin{pmatrix} \frac{\partial^2 E^T}{\partial w_0^2} & \frac{\partial^2 E^T}{\partial w_0 \partial w_1} & \cdots & \frac{\partial^2 E^T}{\partial w_0 \partial w_N} \\ \frac{\partial^2 E^T}{\partial w_1 \partial w_0} & \frac{\partial^2 E^T}{\partial w_1^2} & \cdots & \frac{\partial^2 E^T}{\partial w_1 \partial w_N} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 E^T}{\partial w_N \partial w_0} & \frac{\partial^2 E^T}{\partial w_N \partial w_1} & \cdots & \frac{\partial^2 E^T}{\partial w_N^2} \end{pmatrix} \tag{4.1.12}$$

Let $\underline{\mathbf{g}}$ denote the gradient vector: $\underline{\mathbf{g}} = \frac{\partial E^T}{\partial \underline{\mathbf{w}}}$ . We recognize that the i-th <u>row</u> in $\underline{\mathbf{H}}$ resembles $\frac{\partial}{\partial w_i}\underline{\mathbf{g}}^\top$.

Therefore,

$$\underline{\mathbf{H}} := \frac{\partial}{\partial \underline{\mathbf{w}}}\underline{\mathbf{g}}^\top = \frac{1}{p}\sum_{\alpha=1}^{p} \frac{\partial}{\partial \underline{\mathbf{w}}}\left(\frac{e^{(\alpha)}}{y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \cdot \frac{\partial y(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}{\partial \underline{\mathbf{w}}}\right)^\top \tag{4.1.13}$$

$$= \frac{1}{p}\sum_{\alpha=1}^{p} \frac{\partial}{\partial \underline{\mathbf{w}}}\left(\underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)}\right)^\top \tag{4.1.14}$$

using $\left(\underline{\mathbf{A}}\ \underline{\mathbf{B}}\ \underline{\mathbf{C}}\right)^\top = \underline{\mathbf{C}}^\top \underline{\mathbf{B}}^\top \underline{\mathbf{A}}^\top$ and $\underline{\mathbf{x}}^{(\alpha)\top}\underline{\mathbf{w}} = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)}$

$$= \frac{1}{p}\sum_{\alpha=1}^{p} \left(\frac{\partial}{\partial \underline{\mathbf{w}}}\, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{x}}^{(\alpha)\top} - \frac{\partial}{\partial \underline{\mathbf{w}}}\Big(\underbrace{\underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)}}_{\text{indep. of } \underline{\mathbf{w}}}\Big)^\top\right) \tag{4.1.15}$$

using the product rule

$$\underline{\mathbf{H}} = \frac{1}{p}\sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{x}}^{(\alpha)\top} \tag{4.1.16}$$

### 4.1.4 Represent the cost, gradient and Hessian in matrix form

Let $\underline{\mathbf{X}}$ represent the observations for all training points:

$$\underline{\mathbf{X}} := \begin{pmatrix} | & | & & | & & | \\ \underline{\mathbf{x}}^{(1)} & \underline{\mathbf{x}}^{(2)} & \cdots & \underline{\mathbf{x}}^{(\alpha)} & \cdots & \underline{\mathbf{x}}^{(p)} \\ | & | & & | & & | \end{pmatrix} \in \mathbb{R}^{2\times p} \tag{4.1.17}$$

and let $\underline{\mathbf{y}}_{\text{True}}$ represent the true labels for all training points:

$$\underline{\mathbf{y}}_{\text{True}} := (y_T^{(1)}, \ldots, y_T^{(p)}) \in \mathbb{R}^{1\times p} \tag{4.1.18}$$

From this follows:

$$E^T_{\text{quadr.}} = \frac{1}{2p} \left\| \underline{\mathbf{w}}^\top \underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}} \right\|_2^2 \tag{4.1.19}$$

$$= \frac{1}{2p} \underbrace{\left( \underline{\mathbf{w}}^\top \underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}} \right)}_{1\times p} \underbrace{\left( \underline{\mathbf{w}}^\top \underline{\mathbf{X}} - \underline{\mathbf{y}}_{\text{True}} \right)^\top}_{p\times 1} \tag{4.1.20}$$

with $\underline{\mathbf{y}}_{\text{True}} = \underline{\mathbf{w}}^{*\top} \underline{\mathbf{X}}$:

$$= \frac{1}{2p} \left( \underline{\mathbf{w}}^\top \underline{\mathbf{X}} - \mathbf{w}^{*\top} \underline{\mathbf{X}} \right) \left( \underline{\mathbf{w}}^\top \underline{\mathbf{X}} - \mathbf{w}^{*\top} \underline{\mathbf{X}} \right)^\top \tag{4.1.21}$$

$$= \frac{1}{2p} \left( \underline{\mathbf{w}} - \mathbf{w}^* \right)^\top \underline{\mathbf{X}} \left( \left( \underline{\mathbf{w}}^\top - \mathbf{w}^{*\top} \right) \underline{\mathbf{X}} \right)^\top \tag{4.1.22}$$

with $(\underline{\mathbf{A}}\, \underline{\mathbf{B}})^\top = \underline{\mathbf{B}}^\top \underline{\mathbf{A}}^\top$:

$$= \frac{1}{2p} \left( \underline{\mathbf{w}} - \mathbf{w}^* \right)^\top \underline{\mathbf{X}}\, \underline{\mathbf{X}}^\top \left( \underline{\mathbf{w}}^\top - \mathbf{w}^{*\top} \right)^\top \tag{4.1.23}$$

$$= \frac{1}{2p} \left( \underline{\mathbf{w}} - \mathbf{w}^* \right)^\top \underline{\mathbf{X}}\, \underline{\mathbf{X}}^\top \left( \underline{\mathbf{w}} - \mathbf{w}^* \right) \tag{4.1.24}$$

$$= \frac{1}{2} \left( \underline{\mathbf{w}} - \mathbf{w}^* \right)^\top \underline{\mathbf{H}} \left( \underline{\mathbf{w}} - \mathbf{w}^* \right) \tag{4.1.25}$$

$$\underline{\mathbf{g}} = \frac{1}{p} \sum_{\alpha=1}^{p} \left( \underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - \underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)} \right) \tag{4.1.26}$$

$$= \frac{1}{p} \sum_{\alpha=1}^{p} \left( \underline{\mathbf{x}}^{(\alpha)}\, \underline{\mathbf{x}}^{(\alpha)\top} \underline{\mathbf{w}} - \underline{\mathbf{x}}^{(\alpha)} y_T^{(\alpha)} \right) \tag{4.1.27}$$

$$= \frac{1}{p} \left( \underline{\mathbf{X}}\, \underline{\mathbf{X}}^\top \underline{\mathbf{w}} - \underline{\mathbf{X}}\, \underline{\mathbf{y}}_{\text{True}}^\top \right) = \underline{\mathbf{H}}\, \underline{\mathbf{w}} - \frac{1}{p} \underline{\mathbf{X}} \underline{\mathbf{y}}_{\text{True}}^\top \tag{4.1.28}$$

stationary point:

$$\underline{\mathbf{g}} \overset{!}{=} \underline{\mathbf{0}} \iff \underline{\mathbf{X}}\, \underline{\mathbf{X}}^\top \underline{\mathbf{w}}^* = \underline{\mathbf{X}}\, \underline{\mathbf{y}}_{\text{True}}^\top \tag{4.1.29}$$

$$\underline{\mathbf{w}}^* = \underbrace{\left( \underline{\mathbf{X}}\, \underline{\mathbf{X}}^\top \right)}_{\substack{\text{assumes} \\ \text{invertibility}}}^{-1} \underline{\mathbf{X}}\, \underline{\mathbf{y}}_{\text{True}}^\top \tag{4.1.30}$$

$\underline{\mathbf{w}}^*$ is a global minimum because

1. it is a unique solution of $\underline{\mathbf{g}} = \underline{\mathbf{0}}$ and
2. $$\underline{\mathbf{H}} = \frac{1}{p}\sum_{\alpha=1}^{p} \underline{\mathbf{x}}^{(\alpha)}\,\underline{\mathbf{x}}^{(\alpha)\top} = \frac{1}{p}\,\underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top} = \underline{\mathbf{C}} \tag{4.1.31}$$

   where $\underline{\mathbf{C}}$ is the *empirical covariance matrix* of centered data which is positive definite for invertible $\underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top}$.

positive definiteness:

$$\forall \underline{\mathbf{z}} \in \mathbb{R}^2: \quad \underline{\mathbf{z}}^{\top}\underline{\mathbf{H}}\,\underline{\mathbf{z}} = \frac{1}{p}\,\underline{\mathbf{z}}^{\top}\underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top}\underline{\mathbf{z}} \tag{4.1.32}$$

$$= \frac{1}{p}\left\|\underline{\mathbf{X}}^{\top}\underline{\mathbf{z}}\right\|_2^2 > 0 \quad \text{iff. } \underline{\mathbf{z}} \neq \underline{\mathbf{0}} \tag{4.1.33}$$

### 4.1.5 Solution with regularization

$L_2$ regularization cost:

$$E^R_{[\underline{\mathbf{w}}]} := \frac{1}{2}\sum_{(i,j,v',v)}\left(\mathrm{w}_{ij}^{v'v}\right)^2 = \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 = \frac{1}{2}\underline{\mathbf{w}}^{\top}\underline{\mathbf{w}} \tag{4.1.34}$$

$$\frac{\partial}{\partial \underline{\mathbf{w}}} E^R_{[\underline{\mathbf{w}}]} = \frac{1}{\cancel{2}}\cdot\cancel{2}\cdot\underline{\mathbf{w}} = \underline{\mathbf{w}} \tag{4.1.35}$$

The regularized cost function:

$$R_{[\underline{\mathbf{w}}]} = \underbrace{E^T_{[\underline{\mathbf{w}}]}}_{\substack{\text{training}\\ \text{error}\\ \text{e.g. quadratic cost}}} + \underbrace{\lambda E^R_{[\underline{\mathbf{w}}]}}_{\substack{\text{regularization}\\ \text{term}}} \overset{!}{=} \min_{\underline{\mathbf{w}}} \tag{4.1.36}$$

where

- $E^R \mathrel{\widehat{=}}$ prior knowledge of solution
- $\lambda$ : regularization parameter

Gradient:

$$\underline{\mathbf{g}}^R := \frac{\partial}{\partial \underline{\mathbf{w}}} R_{[\underline{\mathbf{w}}]} = \frac{\partial}{\partial \underline{\mathbf{w}}} E^T_{[\underline{\mathbf{w}}]} + \frac{\partial}{\partial \underline{\mathbf{w}}} E^R_{[\underline{\mathbf{w}}]} \tag{4.1.37}$$

$$= \frac{1}{p}\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top}\underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}^{\top}_{\text{True}}\right) + \underline{\mathbf{I}}_N\,\lambda\,\underline{\mathbf{w}} \tag{4.1.38}$$

$$\Longleftrightarrow \quad \underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top}\underline{\mathbf{w}} - \underline{\mathbf{X}}\,\underline{\mathbf{y}}^{\top}_{\text{True}} + p\lambda\,\underline{\mathbf{I}}_N\,\underline{\mathbf{w}} \overset{!}{=} \underline{\mathbf{0}} \tag{4.1.39}$$

$$\underline{\mathbf{w}}^* = \underbrace{\left(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^{\top} + p\lambda\,\underline{\mathbf{I}}_N\right)}_{\substack{\text{positive definite and}\\ \text{therefore invertible}}}^{-1}\underline{\mathbf{X}}\,\underline{\mathbf{y}}^{\top}_{\text{True}} \tag{4.1.40}$$

## 4.2 Gradient methods

### 4.2.1 The need for gradient-based optimization

We have some cost function we want to minimize but solving the problem analytically is not applicable.

Q 4.2.1: *In what case(s) do we avoid an analytical solution?*

- We cannot fit all the data into memory.
- It involves finding an inverse of a matrix, that is not invertible (e.g. linear regression $(\underline{\mathbf{X}}\,\underline{\mathbf{X}}^\top)^{-1}\underline{\mathbf{X}}\,\underline{\mathbf{y}}_{True}^\top$.
- There is no closed-form solution for the model we're using (e.g. MLP)
- Computing the Hessian is too costly. Newton's method also not applicable.[1]
- Non-stationary data. We cannot easily adapt to changes over time.

We therefore opt for Gradient-based optimization which is an iterative optimization procedure. In the case of a *minimization* problem one can reduce some cost function $E^T_{[\underline{\mathbf{w}}]}$ by taking steps in the direction of steepest *descent*. For minimizing some cost function $E^T_{[\underline{\mathbf{w}}]}$ w.r.t. to a set of weights $\underline{\mathbf{w}}$, it follows that, if

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \underbrace{\eta_t \left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_t}}_{\text{update } \Delta\underline{\mathbf{w}}} \tag{4.2.1}$$

and a small enough $\eta_t \in \mathbb{R}^+$, then the cost at the next step $t+1$ will become less or remain unchanged. That is $E^T_{[\underline{\mathbf{w}}_{t+1}]} \leq E^T_{[\underline{\mathbf{w}}_t]}$.

We refer to $\eta_t$ as the learning rate. It allows us to control the size of the step we take in the direction of the gradient. *Subtracting* the gradient from the current $\underline{\mathbf{w}}_t$ lets us move against the slope in towards some minimum.

The behavior of gradient descent can be controlled by modulating

- The "scope" of the gradient. What do we use for deciding on the direction of the update?
- The learning rate schedule. How big of a step do we take in the direction of the gradient?

### 4.2.2 "Scope" of the gradient

The scope of the gradient method refers to the amount of data used for deciding on the direction of the gradient.

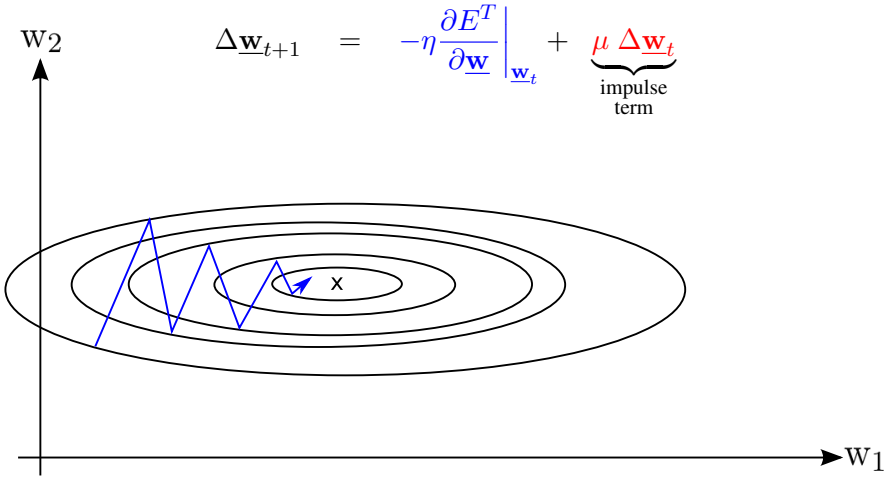
Q 4.2.2: *Which of these can handle non-stationary data?*

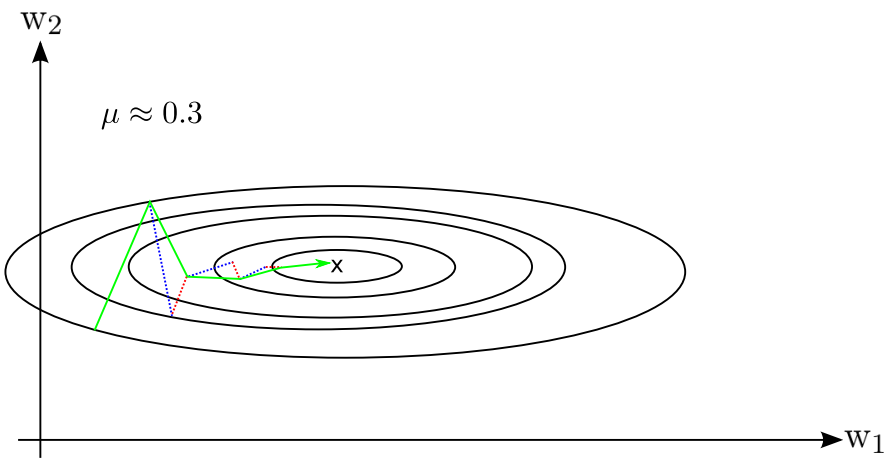
| | | |
|---|---|---|
| batch | $\dfrac{\partial E^T}{\partial \underline{\mathbf{w}}} = \dfrac{1}{p}\sum_{\alpha=1}^{p} \dfrac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}$ | |
| mini-batch | $\dfrac{\partial E^T}{\partial \underline{\mathbf{w}}} \approx \dfrac{1}{\lvert\mathcal{M}\rvert}\sum_{\beta\in\mathcal{M}}^{\lvert\mathcal{M}\rvert} \dfrac{\partial e^{(\beta)}}{\partial \underline{\mathbf{w}}}$ <br> $\lvert\mathcal{M}\rvert \ll p$ <br> where $\mathcal{M}$ is a set of indices sampled randomly from $\{1,\ldots,p\}$ | noisy estimate of $E^T$, becomes less noisy with larger $\lvert\mathcal{M}\rvert$ |
| online | $\dfrac{\partial E^T}{\partial \underline{\mathbf{w}}} \approx \dfrac{\partial e^{(\beta)}}{\partial \underline{\mathbf{w}}}$ <br> a single random data point $(\underline{\mathbf{x}}^{(\beta)}, \underline{\mathbf{y}}_T^{(\beta)})$ | very noisy estimate of $E^T$ |

### 4.2.3 Learning rate schedules

Impulse terms:

$$\Delta\underline{\mathbf{w}}_{t+1} = -\eta \left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_t} + \underbrace{\mu\,\Delta\underline{\mathbf{w}}_t}_{\text{impulse term}}$$

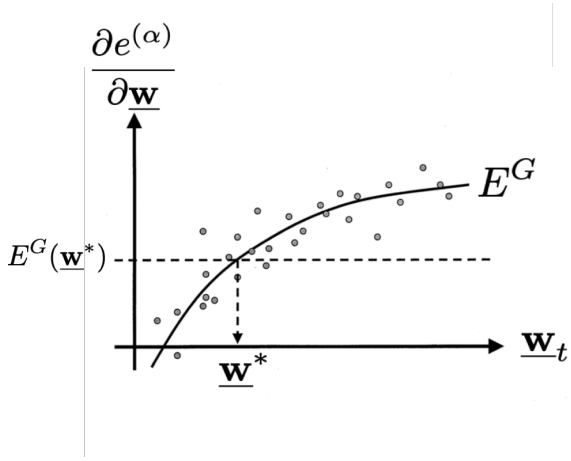


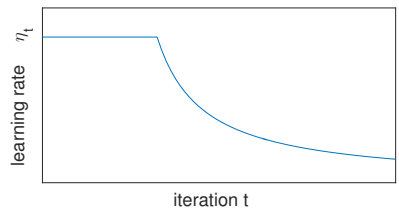

Adaptive step size:

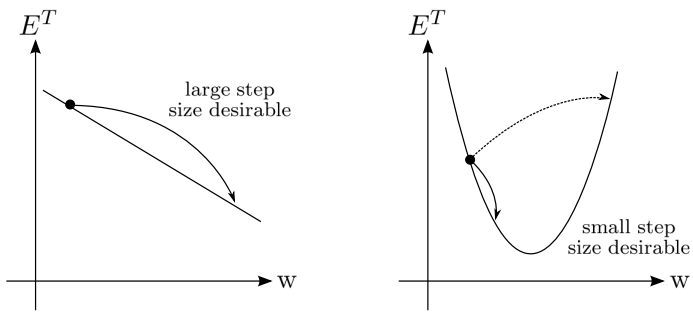
[1] Newton's method finds the root of a function iteratively using the second-order Taylor expansion of the cost function around a point $\underline{\mathbf{w}}_t$. For more details see Haykin Ch. 3.3.

(a) Stochastic Approximation
Robbins and Monro (1951)




(b) Decaying learning rate

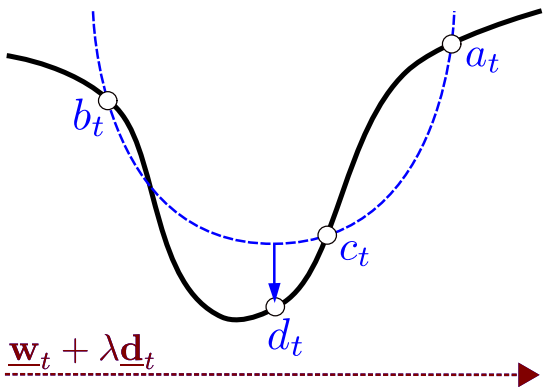



$$\eta_{t+1} = \begin{cases} \rho\eta_t, & \text{if } \Delta E^T < 0, \quad \text{increase step size, if } E^T \downarrow \\ \delta\eta_t, & \text{if } \Delta E^T > 0, \quad \text{decrease step size, if } E^T \uparrow \end{cases}$$

typical values: $\rho = 1.1, \delta = 0.5$

<u>Line search</u>:



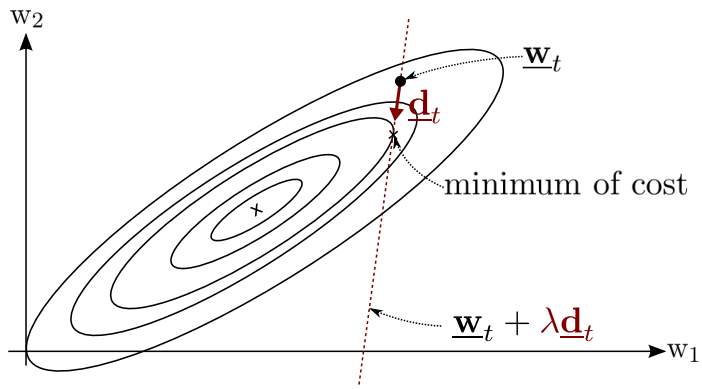
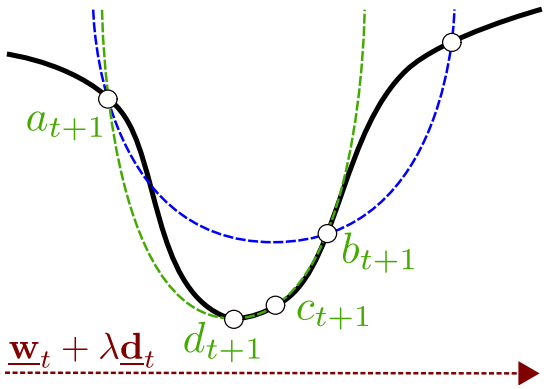



Figure 4.2.2: Line Search




**Line search: successive parabolic interpolation**

**Initialization:** $a_0, b_0, c_0$ (on $\underline{\mathbf{w}}_t + \lambda\underline{\mathbf{d}}_t$); $E^T_{(a_0)}$, $E^T_{(b_0)} > E^T_{(c_0)}$

**while** *stopping criterion not fulfilled* **do**

> Fit a parabola through the three points $a_t, b_t, c_t$ [2mm] Calculate location $d_t$ of its minimum[0mm] Set $c_{t+1} = d_t$, $b_{t+1} = c_t$, $a_{t+1} = \begin{cases} a_t, & E^T_{(a_t)} < E^T_{(b_t)} \\ b_t, & \text{else} \end{cases}$

**end**

For details and implementation see e.g. Numerical Recipes, 2nd edition, Chapter 10.2.

Q 4.2.3: *What are potential pitfalls of parabolic interpolation?*

Compute optimal step size:

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \eta_t \underline{\mathbf{g}}_t, \qquad \text{with optimal step size} \qquad \eta_t = \frac{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{H}}\, \underline{\mathbf{g}}_t}. \quad (4.2.2)$$

### 4.2.4 The Conjugate gradient method

#### 4.2.4.1 Motivation

We have seen how line search allows us to reach the minimum along the direction of the gradient $\underline{\mathbf{g}}_t$ computed at time step $t$. It achieves this by manipulating the magnitude of the step we take towards a lower cost. Looking at what happens over multiple time steps we observe the following movements on the surface of the cost function:

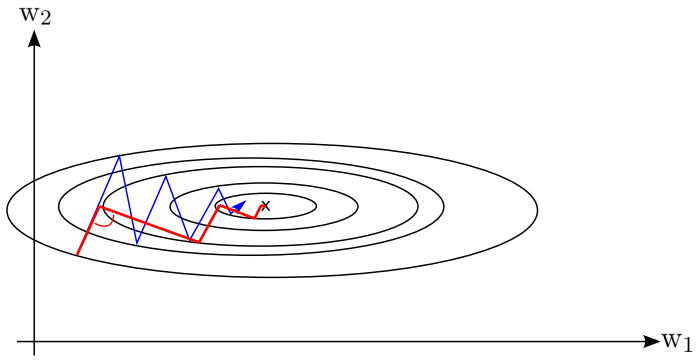



Figure 4.2.3: Weight updates by following line search (red).

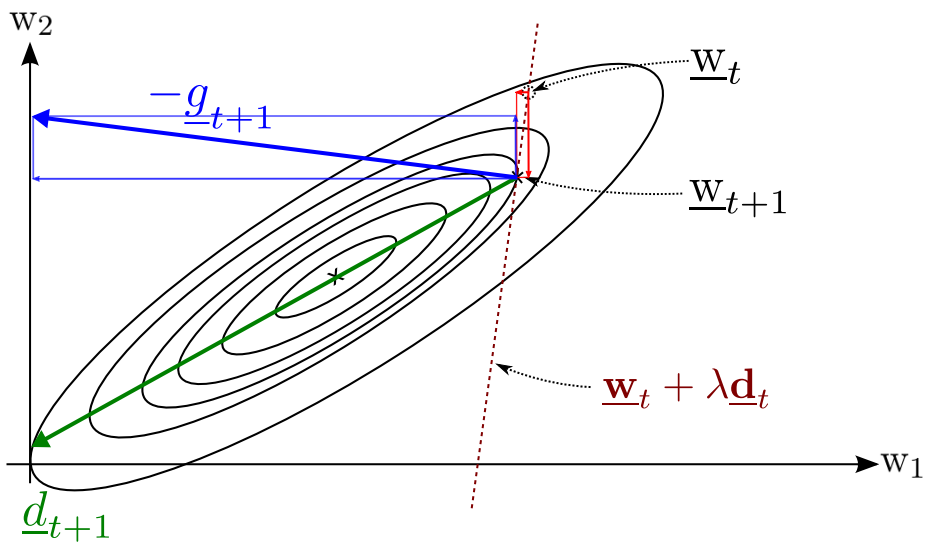



Figure 4.2.4: The conjugate direction avoids undoing progress made in the previous step.

Conjugate gradient method:

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t + \eta_t \, \underline{\mathbf{d}}_t \tag{4.2.3}$$

Don't undo progress of previous search directions $\underline{\mathbf{d}}_t$, $\underline{\mathbf{d}}_{t-1}$.

In other words:

The component of the gradient parallel to the old direction $\underline{\mathbf{d}}_t$ should remain zero along the new direction $\underline{\mathbf{d}}_{t+1}$

**Initialization:** $\underline{\mathbf{w}}$, $\underline{\mathbf{d}} = -\underline{\mathbf{g}}$

**while** *stopping criterion not fulfilled* **do**

> Minimize $E^T$ along $\underline{\mathbf{d}}$ using line-search $\rightarrow$ new $\underline{\mathbf{w}}$ [1mm] Calculate the new conjugate direction $\rightarrow$ new $\underline{\mathbf{d}}$

**end**

$$\underline{\mathbf{d}}_{t+1} = -\overbrace{\left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_{t+1}}}^{\underline{\mathbf{g}}_{t+1}} + \beta_t \, \underline{\mathbf{d}}_t \qquad \text{(conjugate direction)}$$

$$\beta_t = \underbrace{\frac{\underline{\mathbf{g}}_{t+1}^\top \left(\underline{\mathbf{g}}_{t+1} - \underline{\mathbf{g}}_t\right)}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}}_{\text{Polak-Ribiere rule}} \qquad \text{(“smart momentum”)}$$

OR

$$\beta_t = \underbrace{-\frac{\underline{\mathbf{g}}_{t+1}^\top \underline{\mathbf{g}}_{t+1}}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}}_{\text{Fletcher-Reeves}} . \tag{4.2.4}$$

- Subsequent search directions lose conjugacy
  - search direction should be reset to the steepest descent direction at least every $N$ iterations (in practice apply heuristic, e.g. reset every 5 steps)

## 4.3 References

## References

Herbert Robbins and Sutton Monro. A Stochastic Approximation Method. *The Annals of Mathematical Statistics*, 22(3):400 – 407, 1951. doi: 10.1214/aoms/1177729586. URL https://doi.org/10.1214/aoms/1177729586.